

Two-Field Physics Is Enough to Represent One-Dimensional LTE Stellar Atmospheres: Learned and Emulator-Independent Initialization

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ABSTRACT

A one-dimensional LTE stellar atmosphere is commonly initialized through six depth-dependent fields. Four of these fields, however, are constrained by the equation of state, opacity, radiative transfer, and hydrostatic equilibrium once the column mass $m(\tau)$ and temperature $T(\tau)$ are specified. We separate the sufficiency of this two-field representation from the harder problem of predicting a useful initial state. Reconstructing the four dependent fields from exact converged (m, T) profiles preserves both the atmospheric structure and the emergent spectrum. On a 60-star development sample, a learned two-field initializer reduces the mean iteration count from 5.59 to 4.05. An independent 200-star test confirms this computational advantage, but also shows larger tail errors in the predicted profiles and spectra and a slightly lower fraction of usable atmospheres than the six-field initializer. We further construct a hydrostatic grey–convective initializer directly from stellar labels, without an atmosphere emulator. Most development stars converge from this physical approximation, but successful calculations require 14.56 iterations on average, substantially more than for the learned start. A subsequent comparison on the same 200 stars shows the same accuracy–cost trade-off. In a separate 8-node, solar-composition MARCS experiment below 4000 K, a temperature sequence reaches a 3500 K dwarf missed by direct initialization, but its 300% all-layer flux imbalance shows that formal convergence does not yet establish a physically acceptable cool atmosphere. Thus (m, T) is sufficient as a physical representation. Learned predictions reach the convergence basin efficiently but are not yet robust across the sampled label range, whereas the emulator-independent construction is physically informative but slower and less reliable.

Key words. stars: atmospheres – methods: numerical – radiative transfer – techniques: spectroscopic – stars: late-type – stars: fundamental parameters

1. Introduction

Large spectroscopic surveys usually rely on grids of pre-computed stellar atmospheres, such as ATLAS (Kurucz 1970; Castelli & Kurucz 2003), MARCS (Gustafsson et al. 2008), or PHOENIX (Hauschildt et al. 1999; Husser et al. 2013). Interpolation within such grids is fast, but its accuracy depends on the density and geometry of the grid (Mészáros & Allende Prieto 2013). Payne Zero (Ting & Kim 2026, hereafter Paper I) instead accelerates the underlying Kurucz calculation, making it feasible to compute an atmosphere for each set of stellar labels. In this regime, the number of iterations needed for convergence becomes an important part of the computational cost.

The iteration count depends on the initial atmospheric structure. Paper I uses a neural network to predict this structure from the stellar labels. However, a model atmosphere is not an arbitrary collection of profiles. At each of 80 depth points, the solver stores column mass m , temperature T , gas pressure P_{gas} , electron density n_e , Rosseland mean opacity κ_R , and radiative acceleration g_{rad} . Once the stellar labels and the profiles $m(\tau)$ and $T(\tau)$ are specified, the equation of state, opacity calculation, radiative transfer, and hydrostatic equilibrium constrain the remaining four fields. Predicting all six profiles independently can therefore introduce combinations that are individually plausible but mutually inconsistent.

This observation motivates a reduced representation in which only $m(\tau)$ and $T(\tau)$ are supplied. The remaining fields are reconstructed with the atmosphere solver’s own microphysics while the two supplied profiles are held fixed. The closure is iterated to self-consistency before the ordinary temperature correction begins. We distinguish the *two-field representation*, which is the pair (m, T) , from the *two-field physical initializer*, which also includes this self-consistent reconstruction and the subsequent atmosphere iteration. The distinction is important: a representation may be sufficient even when an approximate predictor does not place every star within the convergence basin.

The same representation also allows a complementary physical construction. We derive $m(\tau)$ and $T(\tau)$ from an Eddington-grey atmosphere, hydrostatic equilibrium, approximate Rosseland opacity, and a Saha-aware convective adjustment. This initializer contains no fitted atmospheric profiles and is independent of the neural atmosphere emulator. Its convergence therefore tests whether the solver can be initialized from a compact physical approximation rather than from an interpolated or learned atmosphere.

Cool M-star atmospheres provide a stringent boundary test because molecular equilibrium, opacity, and convection strongly shape their thermal structure (Gustafsson et al. 2008; Husser et al. 2013). The two-field representation also suggests a physical continuation experiment: a converged warmer atmosphere can supply (m, T) for a nearby effective temperature, after which

the dependent fields are reconstructed before the next solve. This tests whether neighbouring cool atmospheres occupy a connected solution branch; it does not extrapolate the learned initializer beyond its training support.

Several lines of related work approach this construction without coinciding with it. Interpolation of precomputed atmosphere grids is the classical way to avoid the cost of a solve, and its error budget in spectra and structure is well characterized (Mészáros & Allende Prieto 2013); more recently, neural interpolation of complete atmosphere structures between grid points reaches high accuracy in the grid’s interior (Westendorp Plaza et al. 2023, iNNterpol), and spectral-level neural emulation of the mapping from labels to spectra underlies The Payne (Ting et al. 2019), on which the fitting side of Paper I builds. Physics-constrained neural surrogates of the atmosphere structure itself, with the governing equations entering the loss, have also been demonstrated (Li et al. 2025, Kurucz-a1, an arXiv preprint and workshop paper). All of these approaches replace or sidestep the solver, whereas here the prediction is used only to start the original solver and is assessed through the subsequent convergence trajectory and final atmosphere.

We address four questions. First, do exact converged (m, T) profiles retain enough information to reconstruct the full atmosphere and its spectrum? Second, can a neural predictor of these two profiles reduce the cost of convergence without compromising the final atmosphere? Third, can an emulator-independent physical approximation reach the same convergence basin? Fourth, can temperature continuation in the reduced coordinates reach selected M-star atmospheres below the 4000 K corpus boundary, and are those atmospheres in flux balance? We separate the representation test from the two initialization tests, and we assess the learned initializer on an independent 200-star sample that was not used during model development.

The remainder of the paper is organized as follows. Section 2 gives the minimal solver description needed to define the reduced state, and Sect. 3 defines the physical reconstruction. Section 4 describes the evaluation design. Section 5 tests representation sufficiency, compares the learned two-field initializer with the six-field reference, evaluates the grey-convective construction, examines the cool M-star boundary, and reports the independent learned test and post-hoc physical comparison. Additional solver diagnostics, the fitted analytic initializer, and reproducibility information are collected in the appendices.

2. Atmosphere solver

2.1. Atmospheric state and physical closure

The reference solver computes a plane-parallel, hydrostatic, one-dimensional LTE atmosphere on a fixed Rosseland optical-depth grid,

$$\tau_{\text{std}}(i) = 10^{-6.875+0.125i}, \quad i = 0, \dots, 80 - 1. \quad (1)$$

It stores six depth-dependent profiles,

$$\mathbf{p} = \{m_j, T_j, P_{\text{gas},j}, n_{\text{e},j}, \kappa_{\text{R},j}, g_{\text{rad},j}\}_{j=1}^{80}, \quad (2)$$

namely column mass, temperature, gas pressure, electron density, Rosseland mean opacity and radiative acceleration. Additional numerical variables and frequency-dependent opacities are retained during iteration but are not part of this six-profile representation.

The closure couples these fields through the equation of state, opacity and radiative transfer. In particular,

$$g_{\text{rad}}(m) = \frac{1}{c} \int_0^\infty \chi_\nu F_\nu d\nu, \quad P_{\text{rad}}(m) = \int_0^m g_{\text{rad}}(m') dm', \quad (3)$$

$$P_{\text{gas}} = g m - P_{\text{rad}}. \quad (4)$$

Here χ_ν is the total extinction per unit mass and F_ν is the monochromatic flux. Turbulent pressure is neglected, while microturbulence enters the opacity calculation as a line-broadening parameter. At fixed labels and fixed (m, T) , the equation of state, opacity and transfer routines therefore determine the other four stored fields. This physical closure is the reason a six-field initializer can contain mutually inconsistent quantities.

2.2. Energy balance

The ordinary solver does more than synchronize the closure: it changes T and m until the total flux is constant,

$$F_{\text{rad}}(m) + F_{\text{conv}}(m) = \sigma T_{\text{eff}}^4, \quad \epsilon_F = \frac{F_{\text{rad}} + F_{\text{conv}}}{\sigma T_{\text{eff}}^4} - 1. \quad (5)$$

We use the temperature correction and damping scheme of Paper I without modification. Each correction also remaps the column-mass coordinate. Consequently, inconsistencies among the initial profiles can alter both $T(\tau)$ and $m(\tau)$ during the first iteration.

2.3. Fixed-point iteration

Writing $\mathbf{x}$ for the complete atmospheric state, one iteration can be summarized as

$$\begin{aligned} \mathbf{x}^{(n)} &\longrightarrow (P_{\text{gas}}, n_{\text{e}}, \chi_\nu)^{(n)} \longrightarrow (g_{\text{rad}}, \kappa_{\text{R}}, F_{\text{conv}})^{(n)} \\ &\longrightarrow (T, m)^{(n+1)} \xrightarrow{\text{remap}} \mathbf{x}^{(n+1)}, \end{aligned} \quad (6)$$

or $\mathbf{x}^{(n+1)} = \mathcal{G}(\mathbf{x}^{(n)})$. The converged atmosphere is the fixed point of this map. Convergence is declared from the deep-layer temperature change,

$$\epsilon_T = \max_{j \in \text{deep}} \left| \frac{T_j^{(n+1)} - T_j^{(n)}}{T_j^{(n+1)}} \right| < 5 \times 10^{-4}, \quad (7)$$

with a minimum of three iterations. All calculations use the convergence criterion and iteration scheme described in Paper I.

3. Two-field physical representation

3.1. Choice of independent profiles

We use (m, T) as the two independent depth-dependent coordinates and treat the remaining fields as functionals of those coordinates, the stellar labels and the fixed depth grid:

$$(P_{\text{gas}}, n_{\text{e}}, \kappa_{\text{R}}, g_{\text{rad}}) = \mathcal{F}[m(\tau), T(\tau); \theta], \quad (8)$$

where θ denotes the stellar labels. The operator $\mathcal{F}$ consists of the same equation of state, opacity calculation, radiative transfer, and hydrostatic closure used during the atmosphere iteration, but is evaluated without changing m or T . Thus two independent profiles determine the four remaining profiles through the physical closure. The six stored profiles therefore occupy

a constrained functional subspace parameterized by two depth-dependent functions, rather than six independently variable functions.

Figure 1 compares three ways to initialize the same solver. The six-field neural initializer predicts all six stored fields. The learned two-field initializer predicts only (m, T) and synchronizes the four dependent fields at fixed coordinates. The hydrostatic grey–convective initializer instead constructs m , T , and an approximate Rosseland opacity directly from the stellar labels. All three initial states then enter the same atmosphere iteration.

3.2. Synchronization to self-consistency

One evaluation of $\mathcal{F}$ is insufficient. The populations and opacities are evaluated at the input gas pressure, while the transfer solution changes the radiation pressure and hence the gas pressure implied by Eq. (4). If the energy iteration starts immediately, its first temperature correction acts on that inconsistency. Even when the exact converged (m, T) profiles are supplied, a single closure evaluation therefore changes the uppermost column-mass layers.

The inconsistency is largest at the top boundary, where optical depth is initialized by

$$\tau_0 = \kappa_0 m_0, \quad (9)$$

which assumes constant opacity above the first grid point. This approximation localizes the largest residual to the surface layers. Repeating the closure reduces the residual substantially, whereas modifying the initial radiation pressure alone does not.

We therefore hold m and T fixed, repeat the four physical stages, and monitor the largest pass-to-pass change in $\log_{10} P_{\text{gas}}$. The exact-coordinate calculations use 3 passes. The learned initializer stops below 1.00×10^{-3} dex, with at most 8 passes. Cases that do not reach this tolerance do not proceed to the atmosphere iteration and are reported as reconstruction failures. The four dependent fields are evaluated before any temperature correction or remapping. Appendix D tests the remaining grid and top-boundary limitations.

4. Atmosphere sample and evaluation

4.1. Atmosphere sample and data splits

All experiments draw on 52 199 converged atmosphere models on the grid of Eq. (1). The five-label support spans 4000–10 500 K in effective temperature, 0.7–5.3 in $\log g$, -2.5 to $+0.5$ in $[M/H]$, -0.1 to $+0.5$ in $[\alpha/M]$, and 0.5 – 4.0 km s^{-1} in microturbulence. The learned model is trained on the corpus training split and assessed on a 60-star development sample excluded from training and validation. A separate 200-star test sample is disjoint from all training, validation, and development rows. It was selected before the final model was evaluated and is an independent test for the learned initializer. The same 200 stars had already been examined before the hydrostatic grey–convective initializer was introduced. Their subsequent use for that method is therefore a post-hoc comparison rather than independent evidence.

The corpus contains only atmospheres that converged in fewer than thirty iterations. Every convergence claim is therefore conditional on the region already represented by this corpus; the experiments do not test whether a new initializer rescues labels for which the solver fails from the outset.

A separate cool-star boundary sample is not drawn from this corpus. It consists of 8 native MARCS atmospheres at solar composition: 4 dwarfs with $\log g = 5.0$ and $\xi = 1 \text{ km s}^{-1}$, and 4 giants with $\log g = 1.5$ and $\xi = 2 \text{ km s}^{-1}$. Each gravity sequence contains $T_{\text{eff}} = 3000, 3300, 3500$, and 3750 K . This sparse grid is designed to contrast high- and low-gravity atmospheres below the 4000 K corpus boundary; it is not a representative sample of the M-star population.

4.2. Initialization methods

We compare three classes of initialization. The Paper I reference model predicts all six profiles directly. The learned two-field model predicts only (m, T) and obtains the remaining profiles from the physical reconstruction of Sect. 3. Its final configuration was selected using the development sample before it was applied to the independent test sample. The grey–convective initializer supplies (m, T, κ_R) from physical approximations with no fitted atmospheric profiles. It is evaluated first on the development sample and then, without changing the construction or numerical limits, on the same 200-star sample. A compact fitted analytic approximation is included as an additional comparison in Appendix E.

For the learned two-field model, reconstruction begins from the stellar labels and $P_{\text{gas}} = gm$; no predicted field from the six-field model is used. This independence applies to inference, not to training: the target (m, T) profiles come from converged full atmospheres. The synchronization criterion is defined through gas pressure, so the dependent fields are self-consistent only to the stated numerical tolerance.

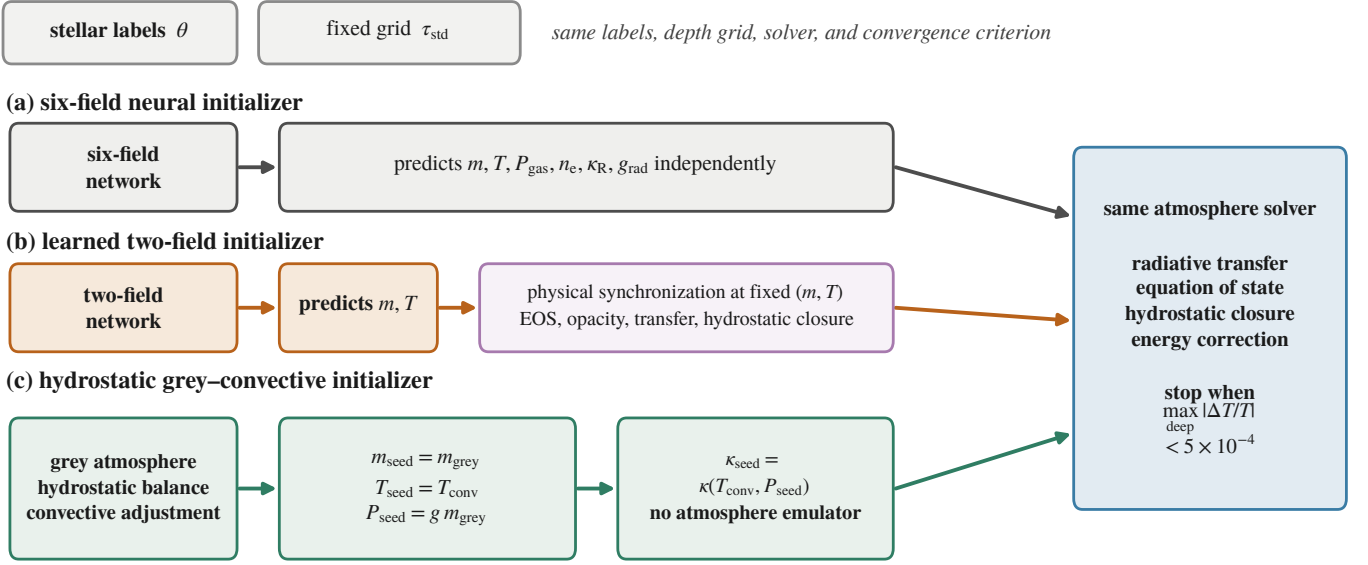
For the physical experiment, the grey, coupled convective, and grey–convective initializers are compared on the same 60 stars. Each calculation is followed for at most 60 iterations and 900 s. These common limits define the reported non-convergences and timeouts. The comparison isolates the effect of retaining the grey column mass when the temperature profile is adjusted for convection.

For the cool-star experiment, each MARCS node is first tested by supplying its native (m, T) profiles directly to the Payne Zero physical closure. A second calculation begins at 4000 K and proceeds through $3750 \rightarrow 3500 \rightarrow 3300 \rightarrow 3000 \text{ K}$. Only (m, T) is transferred between neighbouring temperatures; the dependent fields are reconstructed at every step. Both calculations use the unchanged convergence criterion and a 30-iteration limit. Carrying the full atmospheric state between steps is retained only as a control and does not determine the reported atmospheres.

4.3. Performance measures

We quantify performance at three levels. Profile accuracy is measured against the converged atmosphere. Numerical behavior is characterized by the convergence fraction, iteration count, residual contraction, and frequency of non-monotonic trajectories. Finally, converged spectra are compared over 400–900 nm at $R = 20\,000$. We retain 5.0×10^{-3} as a common reference level across the experiments. The reference solver can itself exceed this value when restarted from a different converged state, so crossing it is reported as a discrepancy rather than treated as a universal pass–fail boundary. Appendix B gives the complete definitions.

For the M-star boundary sample, no spectrum is yet available. We therefore report the maximum absolute relative flux error over all depth layers at the final solver iteration. This quantity



The convective temperature correction does not trigger a second integration of the column-mass coordinate.

Fig. 1. Three initialization paths for the same atmosphere solver. (a) The six-field network predicts all six stored fields directly. (b) The two-field network predicts only $m(\tau)$ and $T(\tau)$. At fixed (m, T) , the equation of state, opacity, radiative transfer and hydrostatic closure are repeated until the pressure is self-consistent. (c) The emulator-independent construction retains the grey-atmosphere column mass, applies a convective temperature adjustment, and evaluates the opacity at $(T_{\text{conv}}, P = g m_{\text{grey}})$. The temperature adjustment does not trigger a second integration of the column-mass coordinate. All three paths use the same stellar labels, depth grid, solver, and convergence criterion.

tests flux balance independently of the temperature-based stopping rule; no post-hoc flux threshold is imposed.

5. Results

The representation and initialization experiments use the 60-star development sample. The independent learned test and the post-hoc physical comparison use the same 200 stars, with the different interpretations established in Sect. 4.

5.1. Representation sufficiency

We first isolate the representation from any prediction error. For each evaluation star, we retain the converged (m, T) profiles, discard the other four fields, and reconstruct them through $\mathcal{F}$.

Figure 4 gives the resulting comparison. The four rebuilt fields agree with the values to which the solver itself converged at 10^{-3} or better in the median. Across the four fields, the point-wise p_{90} errors range from 3.23×10^{-3} to 5.66×10^{-3} . The largest individual errors, 1.36×10^{-1} – 3.60×10^{-1} , are confined to the extreme upper and lower boundary layers.

We next restart the solver from the reconstructed atmosphere and, as a reference, from the original six-field converged state. Both initializations converge for all stars. The reconstructed states require 3.37 iterations on average, compared with 3.23 for the six-field states. 55 of 60 reconstructed atmospheres reach the three-iteration minimum, compared with 57 for the reference. Only 5 stars differ in iteration count, by no more than 2 iterations.

Because most stars reach the minimum number of iterations, the small difference between 3.37 and 3.23 is not a meaningful measure of convergence speed. Both initial states also begin at a first-iteration residual of about 4.20×10^{-4} , already inside

the 5×10^{-4} convergence threshold. Their contraction ratios and non-monotonic fractions therefore characterize numerical fluctuations near the fixed point and are not directly comparable to initializations that begin farther from convergence.

We finally test whether the reconstructed state changes the emergent spectrum. Comparing the converged spectra obtained from the reconstructed and six-field states over 400–900 nm, we find that the normalized flux differs by a median of 8.28×10^{-5} , with a maximum of 1.39×10^{-3} and 0 of 60 stars above the common reference level; the median is roughly 60 times smaller than that level. Total flux and continuum behave in the same way (Table 5, first row). Figure 2 shows this comparison directly across the label support, including the star with the largest normalized-flux difference. Within the tested parameter and wavelength ranges, the four dependent fields can therefore be reconstructed from (m, T) without a measurable change in the emergent spectrum.

5.2. Temperature continuation into the M-star regime

Within this sparse grid, direct starts show a clear gravity dependence (Fig. 3 and Table 1). All 4 of 4 M-giant atmospheres satisfy the unchanged temperature criterion, whereas only 1 of 4 M-dwarf atmospheres does so. The successful dwarf is the warmest node, at 3750 K.

Temperature continuation changes the numerical boundary for the dwarfs. It reaches the 3500 K dwarf after the direct start at the same temperature has exhausted the iteration limit. The sequence then fails at 3300 K and therefore does not reach 3000 K. For the giants, continuation reaches 3750 and 3500 K but fails at 3300 K, while all four direct starts converge. Continuation therefore follows one additional sampled point on the high-gravity cool-star branch, but it is not a general improvement over direct initialization.

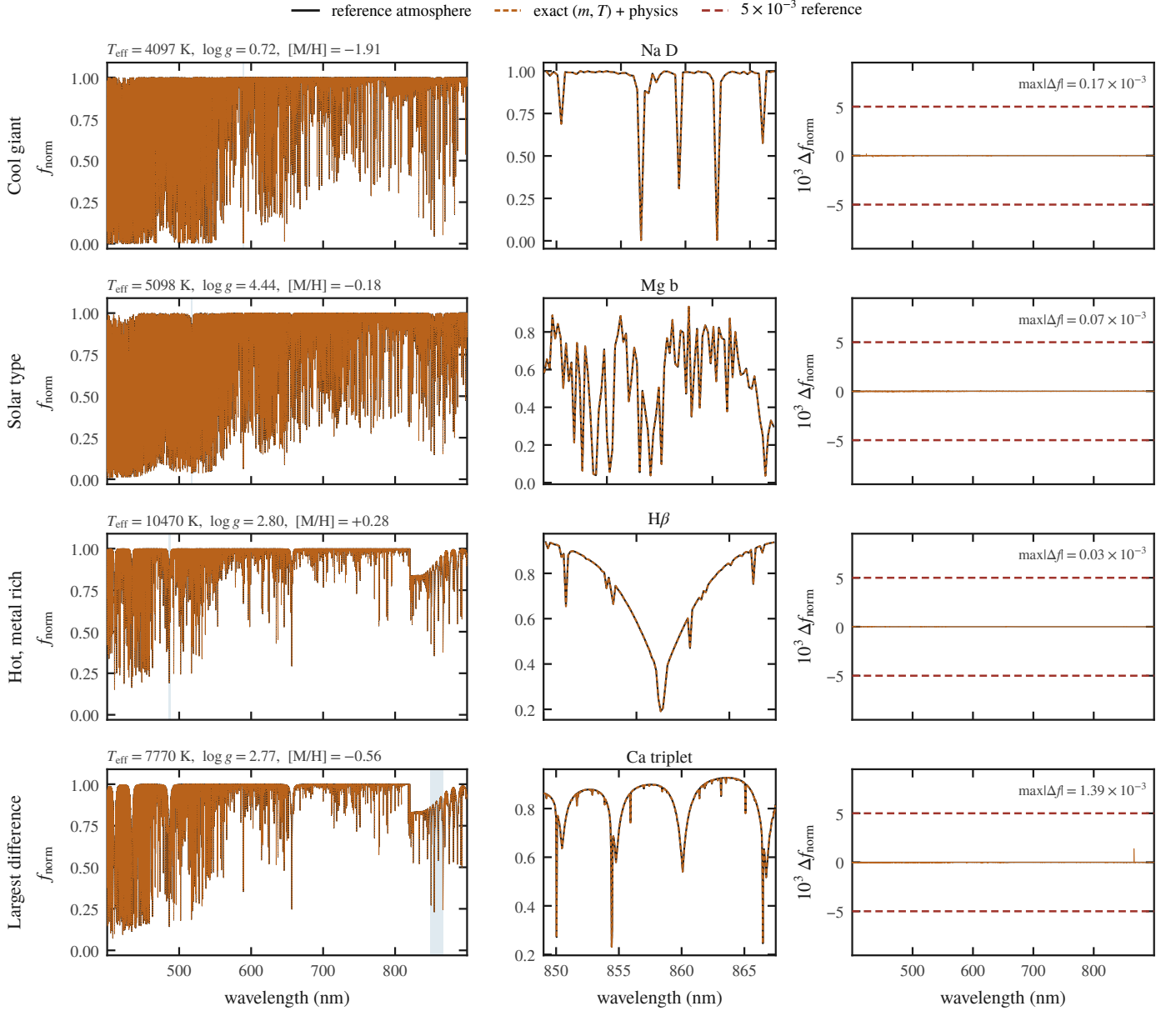


Fig. 2. Representative converged spectra for the exact two-field representation test. The black curves start from the reference six-field atmosphere; the orange curves start after discarding four fields and rebuilding them from the same exact (m, T) coordinates through the solver’s physical closure. Both initial states are then converged with the same atmosphere solver and synthesized identically. The left column shows the full 400–900 nm interval, the middle column resolves one diagnostic feature, and the right column shows the normalized-flux difference relative to the 5.0×10^{-3} reference level. The rows span the tested label support; the bottom row is the largest normalized-flux difference among all 60 stars.

Formal convergence is not the same as a physically satisfactory atmosphere. The continued 3500 K dwarf has a maximum all-layer flux imbalance of 300%, compared with 19.7% at 3750 K. The retained M-giant atmospheres span 1.05–9.49%. The cool-dwarf result therefore demonstrates a numerically connected path, not a validated extension of the atmosphere grid. Spectral comparisons are still absent, and these eight solar-composition nodes do not establish general M-dwarf or M-giant support below 4000 K.

5.3. A learned two-field emulator

We next test the complete two-field physical initializer rather than the representation alone. Its network maps the five stellar labels to $(\log_{10} m, \log_{10} T)$ on the optical-depth grid; the physical

completion then supplies the other four fields before the atmosphere iteration.

Column mass must increase monotonically with depth. We impose this condition through the output parameterization. On the uniform $\log_{10} \tau$ grid, the network predicts an anchor and positive increments,

$$\ell_0 = a(\theta), \quad \Delta \ell_j = \epsilon + \text{softplus}(s_j(\theta)), \quad \ell_j = \ell_0 + \sum_{k=1}^j \Delta \ell_k, \quad (10)$$

where $\ell_j = \log_{10} m_j$, $\epsilon = 10^{-6}$ dex per layer, and the network predicts a and s_j from the stellar labels. The cumulative sum guarantees a strictly increasing column-mass profile.

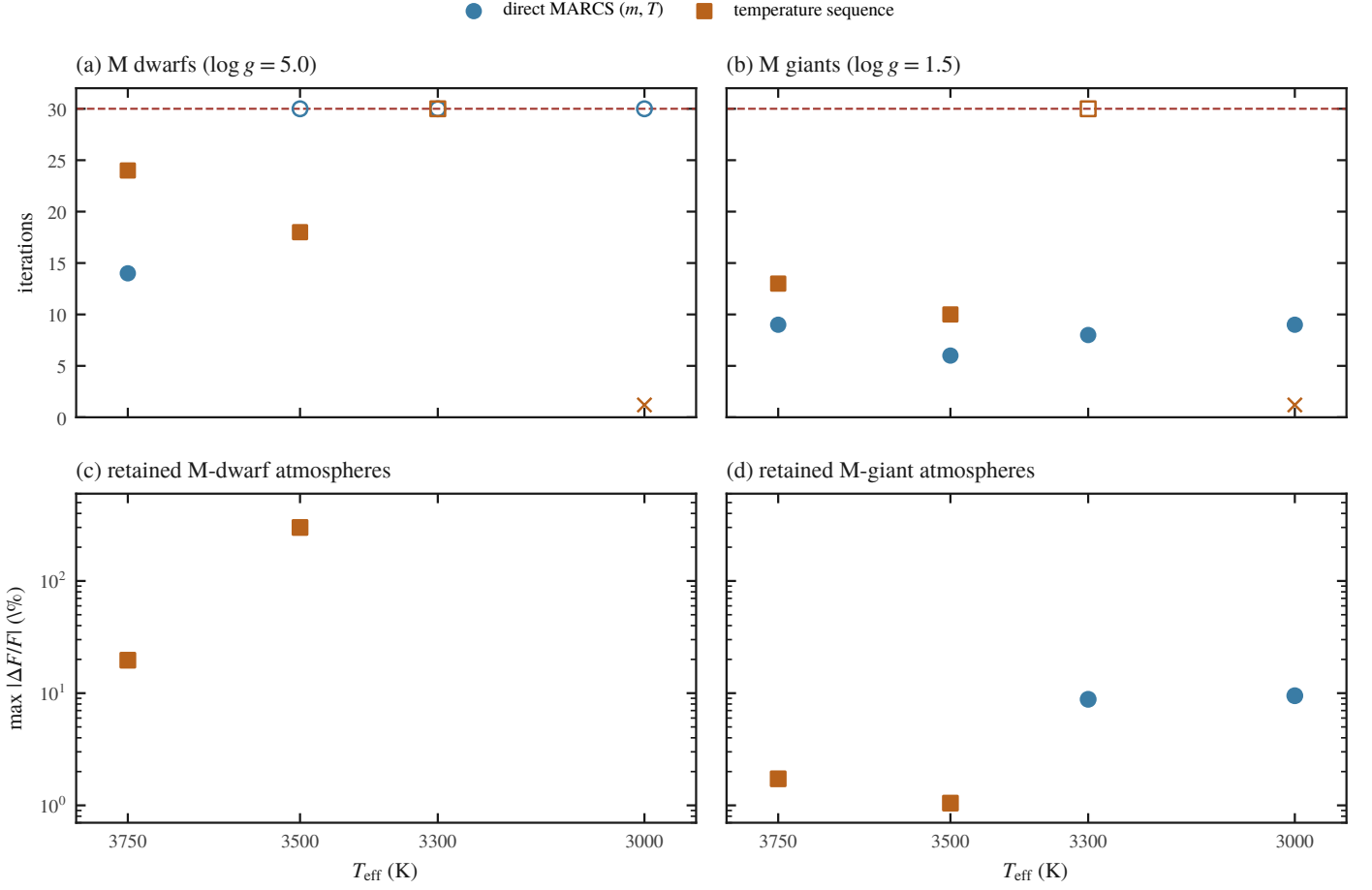


Fig. 3. Temperature and gravity dependence of the cool-star boundary test. The left column shows M dwarfs and the right column M giants. The upper panels compare direct MARCS (m, T) starts with the temperature sequence; filled symbols satisfy the unchanged convergence criterion, open symbols reach the 30-iteration limit, and crosses mark temperatures not reached after an earlier failure in the sequence. The lower panels show the maximum all-layer flux imbalance for the retained atmosphere, using the temperature sequence where it converges and the direct start otherwise. The sequence reaches the 3500 K dwarf missed by the direct start, but the large flux residual prevents interpreting that point as a validated atmosphere.

Table 1. Temperature continuation across eight solar-composition MARCS M-star atmospheres. Direct-start and temperature-sequence entries give convergence status with the iteration count in parentheses; an em dash means that the sequence was not continued past a failed warmer node. The final column gives the maximum absolute all-layer flux imbalance for the retained atmosphere, using the temperature sequence where it converges and the direct start otherwise. The large residual at 3500 K prevents interpreting the formally converged dwarf as a validated atmosphere.

Class	T_{eff}	$\log g$	ξ	Direct start	T_{eff} sequence	$\max \Delta F $ (%)
M dwarf	3750	5.0	1.0	Yes (14)	Yes (24)	1.97×10^1
M dwarf	3500	5.0	1.0	No (30)	Yes (18)	3.00×10^2
M dwarf	3300	5.0	1.0	No (30)	No (30)	---
M dwarf	3000	5.0	1.0	No (30)	---	---
M giant	3750	1.5	2.0	Yes (9)	Yes (13)	1.72×10^0
M giant	3500	1.5	2.0	Yes (6)	Yes (10)	1.05×10^0
M giant	3300	1.5	2.0	Yes (8)	No (30)	8.82×10^0
M giant	3000	1.5	2.0	Yes (9)	---	9.49×10^0

Two learned configurations enter the experiments. The development comparison uses the 4×512 monotone network directly. Before the independent test, the same functional form was re-trained with all development and independent-test rows excluded and was augmented by two label-local corrections. For standardized labels $\mathbf{z}$, each correction uses radial-basis weights

$$w_i(\mathbf{z}) = \exp \left[-\frac{\|\mathbf{z} - \mathbf{c}_i\|^2}{2\sigma^2} \right], \quad (11)$$

and forms depth-dependent logarithmic offsets $\delta_{T,j} = \sum_i w_i a_{ij}^T$ and $\delta_{d,j} = \sum_i w_i a_{ij}^d$. If $d_0 = m_0$ and $d_j = m_j - m_{j-1}$, the corrected profiles are

$$T'_j = T_j 10^{\delta_{T,j}}, \quad d'_j = d_j 10^{\delta_{d,j}}, \quad m'_j = \sum_{k=0}^j d'_k. \quad (12)$$

The weights are not normalized; the corrections therefore decay to zero away from their centres rather than forming a partition of

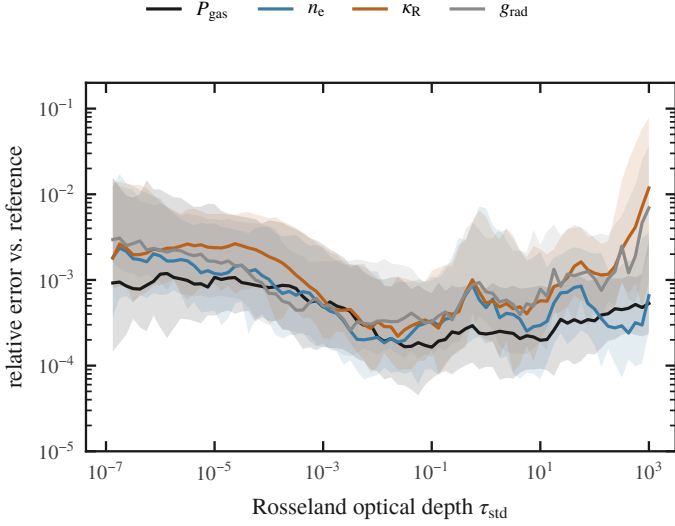


Fig. 4. Relative errors of the four dependent fields reconstructed from the exact converged (m, T) profiles on the 60-star development sample. Curves show medians by depth; shaded regions extend from the 25th to the 90th percentile. This experiment isolates representation sufficiency from initialization error.

unity. The first correction has three centres, $\sigma = 0.30$, and off-sets bounded by 0.2 dex. Its loss is the mean squared logarithmic error in T and m after one differentiable solver iteration relative to the converged six-field atmosphere, plus 0.05 times the corresponding error of the corrected initial profiles relative to the corpus atmosphere. The second has 20 centres and $\sigma = 0.08$; these centres are the remaining profile outliers in the development sample and a previously opened 200-star audit sample. The independent 200-star sample was excluded from both corrections. The base network, both corrections, the independent sample, and all comparison rules were fixed before that sample was examined.

The two-field emulator is more accurate than the six-field network in the median for the shared coordinates, although the comparison inverts in the tail: its p_{95} temperature error is 3.74×10^{-3} against 4.02×10^{-3} , and p_{95} column-mass error 1.52×10^{-2} dex against 2.39×10^{-2} dex.

The more consequential comparison concerns the four fields the two-field network never predicts. Table 2 compares three constructions on the same development sample: six fields predicted directly, learned (m, T) followed by physical completion, and exact (m, T) followed by the same completion. Conditional on successful reconstruction, the learned two-field path reduces the dependent-field errors by factors of 2.5 to 5.7 relative to direct six-field prediction. The improvement arises from enforcing the equation of state, opacity, radiative-transfer, and hydrostatic relations during initialization.

The exact-coordinate calculation quantifies the numerical error of the physical reconstruction; the difference from the learned calculation is due to errors in the predicted (m, T) profiles.

Table 3 and Fig. 5 give the comparison most relevant for initialization. Against the six-field initializer on the same stars and with the same convergence criterion, the learned two-field start reduces the mean iteration count from 5.59 to 4.05 — a reduction of 28% — improves the contraction ratio from 0.656 to 0.547, and cuts the fraction of non-monotonic trajectories from 53.3% to 18.6%. All three initial states in that table begin at a comparable distance from the fixed point, so unlike the exact-coordinate restarts, their contraction statistics are directly comparable.

The learned initializer fails for 4 stars against the six-field initializer’s 1 on the same 60 stars. Of these, 1 do not reach a self-consistent reconstructed state and 3 do not converge after initialization. The exact-coordinate experiment shows that these cases arise from the predicted profiles rather than from an intrinsic insufficiency of the two-field representation.

The direct-output ablation is summarized in Appendix E.

5.4. A hydrostatic grey–convective initializer

The learned experiment does not establish that an emulator is necessary. To test that question directly, we construct $m(\tau)$ and $T(\tau)$ from the stellar labels and physical approximations. The construction combines a grey column-mass profile with a convectively adjusted temperature profile. It first adopts the Eddington-grey relation (Mihalas 1978; Hubeny & Mihalas 2014)

$$T_{\text{grey}}^4(\tau) = \frac{3}{4} T_{\text{eff}}^4 \left(\tau + \frac{2}{3} \right) \quad (13)$$

and integrates

$$\frac{dm_{\text{grey}}}{d\tau} = \frac{1}{\kappa_{\text{R}}^{\text{app}}(T_{\text{grey}}, P_{\text{grey}})}, \quad P_{\text{grey}} = g m_{\text{grey}}. \quad (14)$$

The approximate opacity is a Rosseland harmonic mean over fixed quadrature nodes. It includes H^- and H I bound-free and free-free absorption, H_2^+ and He^- continua, electron scattering, and H I Rayleigh scattering; the electron density is obtained from Saha ionization of hydrogen and the principal electron donors. Published threshold cross-sections are used for the lowest H I levels in the cool branch. This opacity is used only to construct the initial state; the atmosphere iteration subsequently uses the solver’s full opacity calculation. A Saha-aware adiabatic adjustment then replaces the temperature in convective layers (Böhme-Vitense 1958). The solver is initialized with

$$m_{\text{seed}} = m_{\text{grey}}, \quad T_{\text{seed}} = T_{\text{conv}}, \quad \kappa_{\text{seed}} = \kappa_{\text{R}}^{\text{app}}(T_{\text{conv}}, P_{\text{grey}}). \quad (15)$$

The grey column mass is retained after the convective temperature adjustment; it is not re-integrated with the modified temperature. All 60 development initial structures are finite and positive, and the opacity is recomputed consistently from T_{conv} and $P_{\text{grey}} = g m_{\text{grey}}$. The construction contains 0 fitted parameters and does not use the atmosphere emulator.

On the development sample, the initializer converges for 54/60 stars within 60 iterations. Among successful calculations, the mean is 14.56 iterations, the median is 14.0, and the maximum is 29. 17 successful stars require more than 15 iterations. The remaining 6 stars do not converge within 60 iterations. Extending those same cases to 100 iterations does not recover any of them; all 6 reach the 3600-s limit. On the 54 stars for which both the physical and six-field starts converge, the corresponding mean costs are 14.56 and 5.46 iterations, with a median physical-minus-reference difference of 9.0 iterations. Thus independence from the emulator is obtained at a substantial and occasionally prohibitive increase in iteration cost.

To identify which part of the physical construction matters, we compare the selected grey–convective start with a grey atmosphere and with a convective construction that re-integrates column mass after changing the temperature. The grey–convective start converges for 54/60 stars, compared with 52/60 for the grey

Table 2. Median relative error of the four dependent fields over the development sample and 80 depth points. The learned row contains the 59 successful physical reconstructions; the remaining 1 case has no complete dependent-field profile. The exact- (m, T) row quantifies the numerical accuracy of the physical reconstruction. All rows refer to the same sampled stars.

Source of the four fields	P_{gas}	n_e	κ_R	g_{rad}
Six fields predicted directly	1.81×10^{-2}	1.30×10^{-2}	1.43×10^{-2}	3.18×10^{-2}
Learned two-field (m, T) + physics	4.51×10^{-3}	5.24×10^{-3}	5.21×10^{-3}	5.59×10^{-3}
Converged (m, T) + physics	4.05×10^{-4}	5.67×10^{-4}	8.74×10^{-4}	8.47×10^{-4}

Table 3. Convergence from three initializations on the same 60 development stars. All begin at comparable initial residuals, so their contraction statistics can be compared directly. The learned two-field model is the 4×512 monotone network described in Sect. 5.3; the independently tested model in Table 6 includes the radial-basis corrections of Sect. 5.3.

Restart from	Conv.	Iterations	p_{90}	p_{99}	q	Non-mono.	First residual
Six-field initializer	98.3%	5.59	8	11.26	0.656	53.3%	2.44×10^{-3}
Learned two-field + physics	93.3%	4.05	6	8.45	0.547	18.6%	2.75×10^{-3}
Six-field, alternate converged start	93.3%	7.45	10	12.00	0.658	78.3%	6.75×10^{-3}

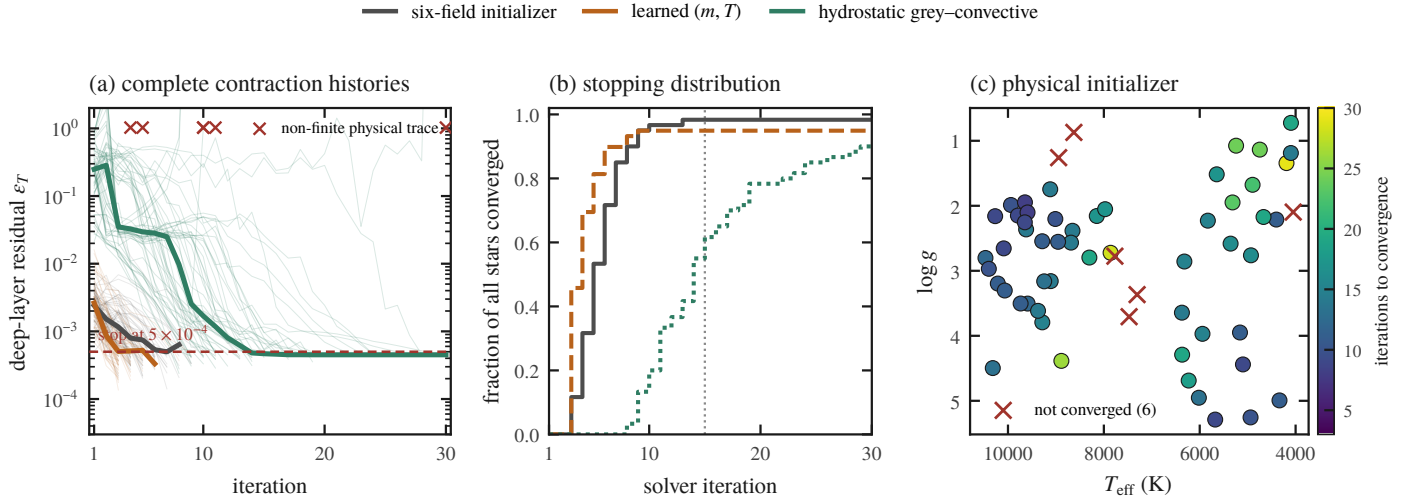


Fig. 5. Convergence on the 60-star development sample. (a) Complete deep-layer temperature-residual histories for the hydrostatic grey-convective initializer (thin green curves), together with thick medians for the six-field, learned two-field, and physical initializers. The black and orange medians use finite residuals from stars still being iterated. For the physical median, each trajectory remains at its last finite residual after it ends; this keeps the 60-star cohort fixed and prevents an apparent late rise caused only by the changing subset. Red crosses along the upper boundary mark the first non-finite residual in each affected physical-initializer history; the horizontal line is the stopping threshold. (b) Fraction of all 60 stars converged by each iteration. Failed stars remain in the denominator, so the physical curve ends at 54/60. (c) Iteration count of the converged physical starts across label space; crosses denote the 6 stars that do not converge within 60 iterations.

start and 49/60 for the coupled convective start. Its gains relative to grey are +2 stars overall and +3 cool stars. The improvement is therefore real but uneven across label space. Retaining the grey mass coordinate is empirically preferable in this comparison, although these convergence counts alone do not identify the physical origin of the difference.

The physical seed is deliberately approximate. Its development-sample median column-mass error is 0.126 dex and its median temperature error is 1.49×10^{-1} . These offsets are visible directly in the profile families of Fig. E.2. They do not prevent most stars from reaching the same atmosphere, but they explain the longer contraction histories in Fig. 5.

Of the 60 development stars, 53 have both a converged physical-start atmosphere and a reference spectrum; 7 do not enter the spectral comparison. The median maximum normalized-flux difference is 3.20×10^{-2} , and 45 paired spectra exceed the common 5.0×10^{-3} reference level. The spectral distribution is

conditional on convergence and is therefore reported separately from the six solver failures.

5.5. Independent learned test and post-hoc physical comparison

The learned model and both local corrections were fixed before the independent 200-star sample was examined, and no subsequent result changed their parameters. The learned initializer does not retain its development profile or spectral accuracy on this sample (Fig. 8). The column-mass p_{95} error more than doubles to 2.66×10^{-2} dex, and 19 stars develop large profile discrepancies (Table 6); 25 of the 189 paired spectra exceed the common reference level.

The learned initializer nevertheless remains faster on the independent sample: the paired calculations save 2.00 iterations in the median, and the bootstrap 95 per cent interval for the mean

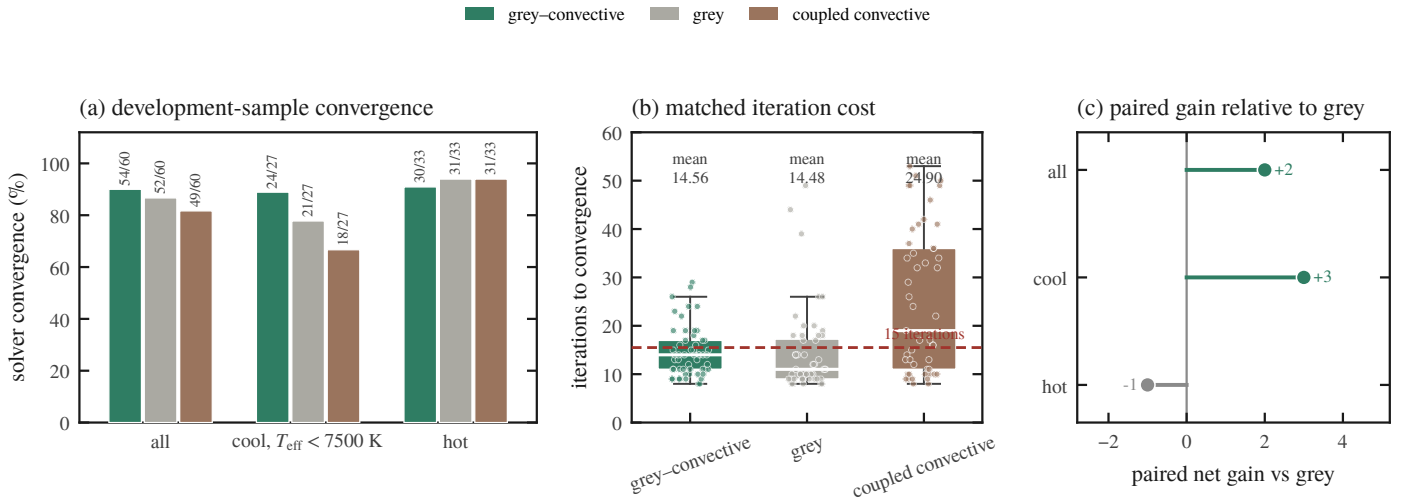


Fig. 6. Comparison of three emulator-independent physical initializations on the 60-star development sample. All calculations use the same stellar labels and numerical settings. (a) Convergence on the full, cool, and hot subsets. (b) Iteration distributions conditional on convergence. The dashed line marks the earlier 15-iteration comparison point. (c) Paired net convergence difference between the grey-convective and grey starts; positive values favour the former.

saving is 1.24 to 1.85. The two local corrections change only 3/200 predictions by more than 10^{-3} dex, so the speed difference is primarily a property of the underlying two-field network.

After the sample had been examined, we repeated the learned calculation with the four dependent fields reconstructed solely from predicted (m , T) and the stellar labels. This removes any dependence on six-field emulator outputs at inference while leaving the convergence and spectral conclusions essentially unchanged (Table 6). Because this reconstruction check was performed after inspecting the sample, it is a post-hoc diagnostic rather than a second independent test.

The hydrostatic grey-convective initializer was also developed after this sample had been examined. Its formulae and numerical limits were fixed before the calculation, and all 200 outcomes are retained. This tests whether the development behavior extends across the same label distribution, but it is not independent evidence.

The physical initializer produces finite, positive seeds for 200/200 stars, but converges less often and requires substantially more iterations than the six-field start (Table 4). The failure set—18 timeouts and 1 calculation that exhausts the iteration budget—is concentrated in temperature: 16 of these 19 cases lie between 7000–9000 K, although this interval contains only 62 of the 200 stars. The failures show no comparable concentration in $\log g$ (Fig. 8c). This temperature dependence is consistent with a limitation near the transition from appreciable surface convection to predominantly radiative transport, rather than with a cool-star or low-gravity boundary alone.

The physical initializer also gives larger conditional spectral differences than either learned comparison (Table 5). Because this distribution contains only stars with both converged atmospheres, it must be read together with the full-sample convergence fraction.

6. Discussion

The exact-coordinate experiment shows that $m(\tau)$ and $T(\tau)$ contain the information needed to reconstruct the remaining stored atmospheric profiles within the numerical precision relevant for the emergent spectrum. At fixed stellar labels, the equation of

state, opacity, radiative transfer, and hydrostatic equilibrium constrain P_{gas} , n_e , κ_R , and g_{rad} . Treating those four quantities as independent coordinates is therefore unnecessary for this atmosphere family and can introduce inconsistencies that the solver must remove.

Prediction is harder than representation. The learned two-field model reduces the iteration count on both the development and independent samples, but its profile and spectral errors have a pronounced independent-sample tail. The two local radial-basis corrections affect only 3/200 independent predictions at the 10^{-3} -dex level; the measured speed improvement is therefore associated mainly with the base two-field model. Faster convergence is not, by itself, evidence of a more accurate final atmosphere. The predictor must be judged through both the solver trajectory and the converged state (Um et al. 2020).

The grey-convective calculation shows that a learned atmosphere is not required to enter the solver basin for most sampled stars. Retaining m_{grey} while applying the convective temperature adjustment gives the highest convergence fraction among the three physical constructions tested. The grey column mass carries the integrated opacity and hydrostatic scaling of the initial atmosphere; re-integrating it after changing T perturbs both coordinates through the approximate opacity. Preserving m_{grey} may therefore retain more of the original hydrostatic structure while the solver adjusts the temperature. This interpretation is plausible but not established by the convergence counts alone. The practical cost is clear: successful development cases require 14.56 iterations on average, and failures persist even when the iteration budget is extended.

The cool-star experiment exposes the distinction between entering a numerical convergence basin and obtaining an atmosphere in radiative-convective balance. Within the eight solar-composition nodes, low-gravity giants are substantially easier to initialize directly than high-gravity dwarfs. This contrast may reflect the stronger coupling of molecular opacity, pressure, and convection in the dense dwarf atmospheres, but the sparse grid cannot identify the mechanism. The successful continuation to 3500 K nevertheless shows that neighbouring solutions in the reduced coordinates can guide the solver where a direct MARCS start fails. Its 300% flux imbalance, and the failure of the next

Table 4. Hydrostatic grey-convective initialization on the development sample and, post hoc, on the same 200-star sample used for the learned test. Every star uses at most 60 iterations and 900 s. Iteration statistics are conditional on convergence; the convergence and failure columns retain the full sample. Spectral conditioning is reported separately in Table 5.

Sample	Converged	Non-conv./timeout/error	Mean iter.	Median	Max	> 15
Development	54/60	6/0/0	14.56	14.0	29	17
Post-hoc 200	181/200	1/18/0	15.69	14.0	47	72

Table 5. Normalized-flux differences between converged atmospheres over 400–900 nm at $R = 20\,000$. The 5.0×10^{-3} value is retained as a common reference level, not as a universal acceptability boundary. Only stars with converged atmospheres from both initial states enter each row. The final row measures the start dependence of the reference solver and provides the numerical scale for the other comparisons.

Comparison	N	Median	Maximum	Above reference
Converged (m, T) vs. reference atmosphere	60	8.28×10^{-5}	1.39×10^{-3}	0/60
Learned two-field vs. six-field reference	56	1.61×10^{-3}	5.04×10^{-3}	1/56
Grey-convective vs. reference, development	53	3.20×10^{-2}	1.33×10^{-1}	45/53
Grey-convective vs. reference, post-hoc 200	179	2.59×10^{-2}	1.70×10^{-1}	150/179
Six-field reference vs. alternate converged start	56	3.44×10^{-3}	1.50×10^{-2}	16/56

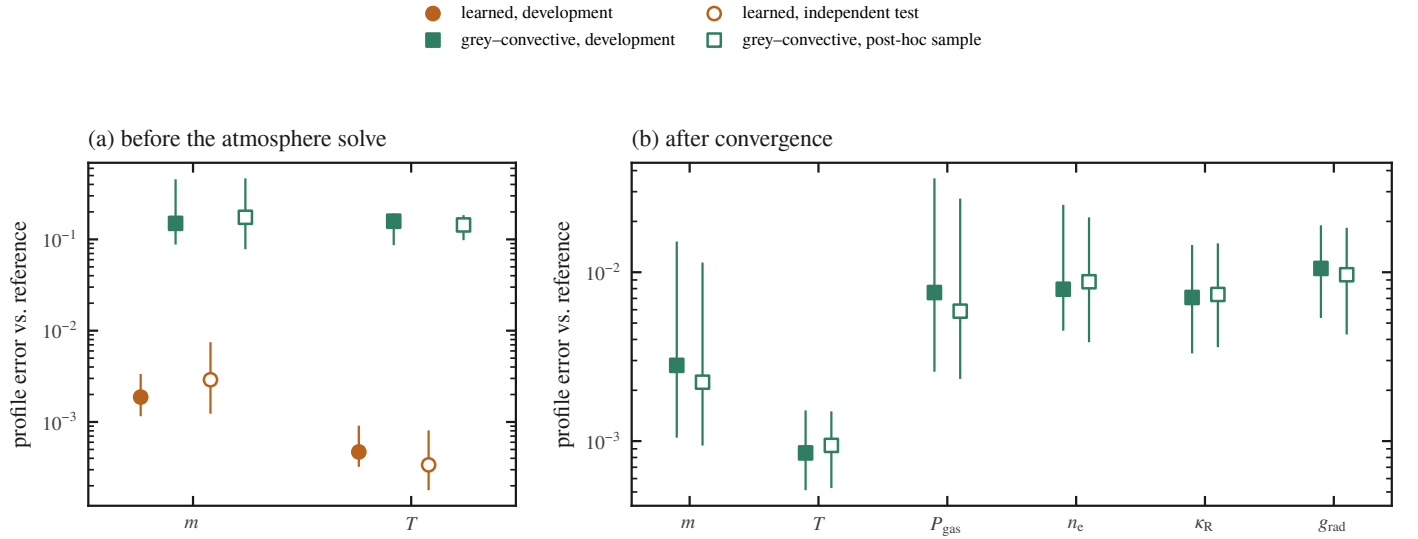


Fig. 7. Profile accuracy of the learned and emulator-independent initializations. (a) Errors in the two independent seed profiles before the atmosphere solve. Throughout the figure, column-mass differences are measured in dex and all other differences are relative; filled symbols denote the development sample and open symbols the 200-star sample. (b) Differences between the converged physical-start atmospheres and the reference atmospheres for all six stored fields. Points are medians over stars of each star’s median layer error; bars span the 16th–84th percentiles. Panel (b) is conditional on convergence and is therefore interpreted together with the full-sample convergence fractions in Table 4. Provisional closure values used before the solve are not treated as accuracy measurements.

temperature step, show equally clearly that reaching the temperature stopping criterion is not enough. A flux-aware convergence test and spectral validation are required before the low-temperature domain can be extended.

The scope remains limited by the atmosphere corpus, which contains models that originally converged in fewer than thirty iterations. The experiments do not show that either initializer recovers regions where the reference method fails. The targeted M-star calculation lies outside that corpus but covers only eight solar-composition nodes and has no spectral validation. Solver iterations also exclude the cost of constructing the initial state, and the spectral comparison is restricted to 400–900 nm. Finally, the 200-star physical comparison was performed after that sample had been used for model testing, and the spectral distributions are conditional on both atmospheres converging. These results establish representational sufficiency and expose

the accuracy–cost trade-off; they do not yet establish robustness for routine use.

The residual reconstruction error is concentrated near the upper boundary. Appendix D shows that neither refinement of the interior optical-depth grid nor denser quadrature removes this behavior, implicating the approximation used above the first depth point. Atmospheres extending to smaller optical depth are needed to determine whether a modified surface boundary condition reduces this residual. For the physical initializer, the next methodological step is to improve the approximate mass-column profile while retaining an independent evaluation sample for subsequent testing.

Table 6. The learned two-field initializer on the development sample and the independent 200-star test, followed by a post-hoc check that rebuilt the same predicted (m, T) profiles without fields from the six-field model. The independent column contains the original independent calculation. The final column changes only the reconstruction seed and is not additional independent evidence. Iteration means for the 200-star columns use stars with both compared atmospheres available.

Quantity	Development	Independent test	Post-hoc reconstruction
Stars	60	200	200
Temperature p_{95}	2.06×10^{-3}	3.38×10^{-3}	same prediction
Column mass p_{95} (dex)	1.19×10^{-2}	2.66×10^{-2}	same prediction
Large profile errors (T/m)	0 / 0	6 / 18	same prediction
Usable two-field atmospheres	60/60	192/200	190/200
Usable six-field atmospheres	60/60	193/200	193/200
Mean two-field iterations	3.77	4.14	4.12
Mean six-field iterations	—	5.68	5.69
Maximum spectral difference	4.47×10^{-3}	1.42×10^{-2}	1.42×10^{-2}
Above spectral reference level	0/60	25/189	25/188

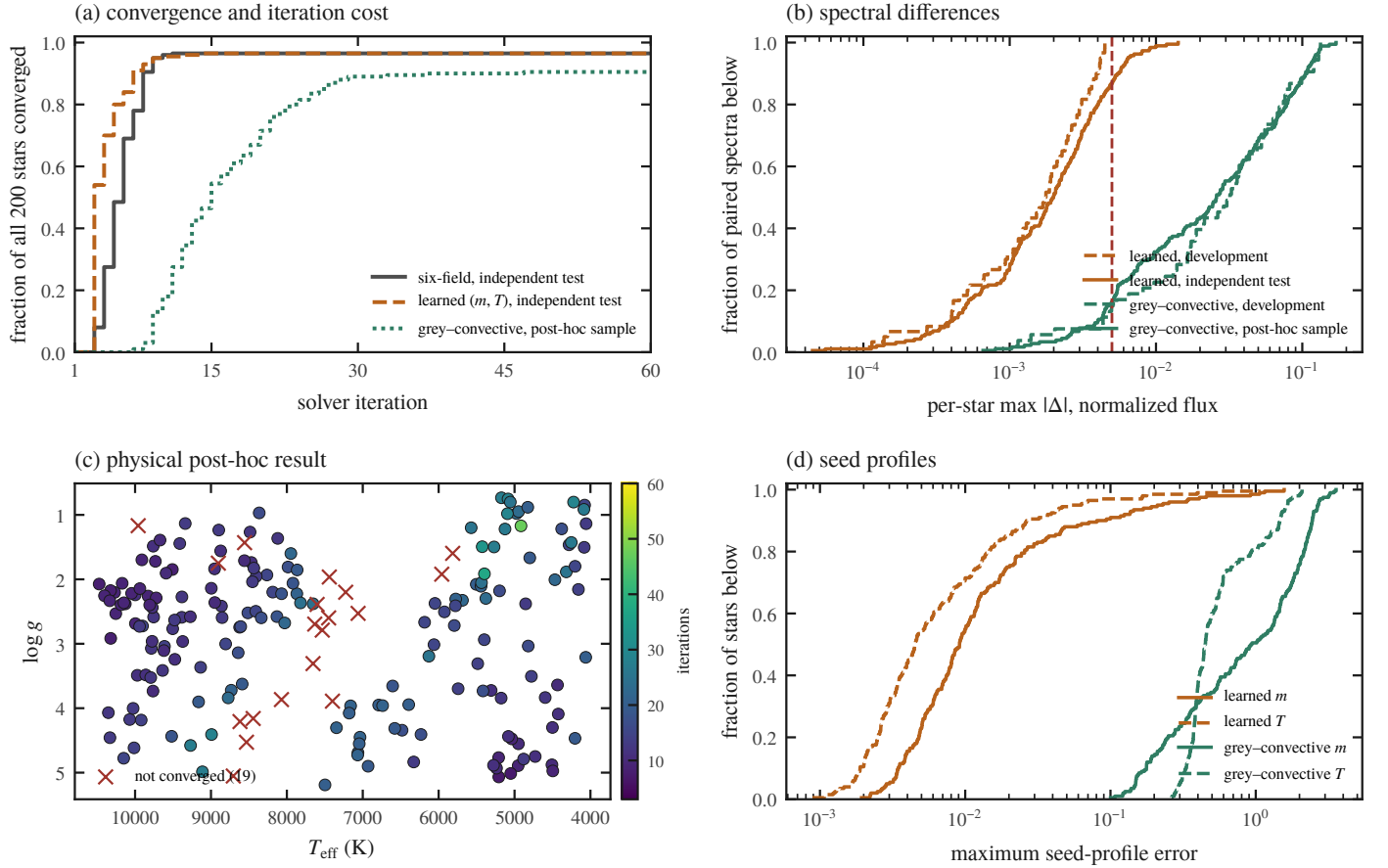


Fig. 8. Learned and physical initialization on the same 200-star sample. The learned calculation is the independent test; the hydrostatic grey-convective calculation is a post-hoc comparison. (a) Fraction of all 200 stars converged by each iteration; failures remain in the denominator. (b) Cumulative normalized-flux differences on the development and 200-star samples. Spectral curves are conditional on both compared atmospheres having converged. (c) Iteration count of the converged physical starts over T_{eff} and $\log g$; crosses mark non-convergeds or timeouts. (d) Maximum pre-solver errors in the seed profiles, measured in dex for m and relatively for T . Sample status, convergence accounting, and spectral conditioning are kept separate.

7. Conclusions

Within the sampled stellar-label range, a converged 1D LTE atmosphere can be represented by $m(\tau)$ and $T(\tau)$. Reconstructing the four dependent profiles through the solver's physical closure reproduces both the atmospheric structure and the emergent spectrum. The closure must be iterated at fixed (m, T) before the temperature correction begins.

A learned two-field initializer reduces the mean iteration count from 5.59 to 4.05 and produces smoother convergence on the development sample. The speed advantage persists on the independent 200-star sample, whereas the profile and spectral accuracy develop a substantial tail. The representation is sufficient, but the present predictor is not uniformly reliable across the sampled label range.

An emulator-independent grey-convective atmosphere initializes most sampled stars successfully, but converged development cases require about 14.56 iterations on average and failures remain. A learned atmosphere is therefore not physically necessary for initialization, but it remains valuable for computational efficiency. The physical route requires an independent evaluation before routine use.

Temperature continuation in the reduced coordinates reaches a 3500 K M dwarf for which direct initialization fails. It does not yet extend the validated physical domain: the sequence fails at the next cooler node, and the recovered atmosphere retains a 300% all-layer flux imbalance. This result motivates continuation as a route into cool-star parameter space while requiring flux balance and spectra before any claim of M-dwarf coverage.

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Appendix A: Physical reconstruction algorithm

Given $m(\tau)$, $T(\tau)$, and labels θ , the dependent profiles are reconstructed on the grid of Eq. (1) as follows.

1. *Seed.* An atmosphere object is constructed carrying m and T on the optical-depth grid. The remaining fields are initialized from the labels and hydrostatic balance alone: $P_{\text{gas}} = g m$, with positive n_e and κ_R placeholders and $g_{\text{rad}} = 0$. The molecular equation of state and the first opacity and transfer passes overwrite these values. They therefore define only the starting point of the physical closure; no fields from the six-field model are used. A profile whose column mass is not strictly increasing is not physically admissible and is excluded.
2. *Populations.* The equation of state is solved for the chemical populations and the electron density at each layer.
3. *Opacity.* Continuum and line opacities are evaluated and combined into the Rosseland mean κ_R .
4. *Transfer.* The radiative transfer solution supplies the radiation pressure and the radiative acceleration g_{rad} .
5. *Closure.* The gas pressure is recomputed from Eq. (4) with the radiation-pressure term now known.

Steps 2–5 form one closure iteration. They are repeated because the populations and opacities in steps 2–3 use the pressure at the start of a pass, while step 5 updates that pressure from the radiation field they imply. The exact-coordinate calculations use 3 fixed iterations. The learned initializer uses at most 8 iterations and a pressure-change tolerance of 1.00×10^{-3} dex. Initial structures that do not reach this tolerance are reported as reconstruction failures. Neither m nor T is modified, and the depth grid is unchanged.

Appendix B: Accuracy and convergence criteria

Profiles. We calculate the pointwise relative error in temperature and the pointwise absolute error in $\log_{10} m$ relative to the converged atmosphere. A star is classified as having a large profile discrepancy if its maximum temperature error exceeds 10 percent or its maximum column-mass error exceeds 0.1 dex.

Convergence. The same atmosphere solver is initialized from each tested state. We record the fraction of stars that converge; the iteration count of the converging trial; the geometric mean of the ratio $q_k = r_{k+1}/r_k$ of successive deep-layer temperature residuals; and the fraction of trajectories on which r_k is not monotonically decreasing. A state that satisfies the temperature stopping rule but is not finite is counted as a non-convergence.

Spectra. Converged atmospheres from each pair of initial states are synthesized over 400–900 nm at $R = 20\,000$ in double precision. Three quantities are compared: the maximum absolute difference in normalized flux; the maximum absolute difference in total flux, divided by the continuum; and the maximum relative difference in the continuum. The common reference level is 5.0×10^{-3} for each quantity. The reference-solver restart comparison in Table 5 provides its numerical scale; it is not treated as a universal boundary between acceptable and unacceptable atmospheres.

Appendix C: Solver behavior by label sampling

Table C.1 shows what the unmodified solver does when started from the six-field initializer. The solver and the initializer are identical in all four rows; only the way in which the stellar labels are drawn changes, and the resulting spread is the point of the

comparison. Drawing labels uniformly across the label ranges the corpus spans gives 85.3% convergence and 14.7% outright failures, whereas resampling the corpus's own label distribution gives 99.2% convergence, with a 90th-percentile iteration count of 8. This contrast quantifies the sensitivity of convergence statistics to the sampled label distribution.

Appendix D: Resolution and boundary controls

The representation-resolution calculation maps converged (m, T) curves through intermediate optical-depth grids before physical reconstruction. Increasing the intermediate grid from 40 to 640 points changes the median column-mass round-trip error from 3.08×10^{-5} to 6.55×10^{-9} , while the solver behavior remains near the three-iteration floor. A separate quadrature control increases the integration density by a factor of 16 while holding the surface seed fixed. Its top-layer p_{99} residual changes only from 0.03571 to 0.03497 dex, and the maximum from 0.66143 to 0.66143 dex. The remaining residual is therefore associated with the upper boundary rather than the interior grid spacing.

Appendix E: Analytic initializer and additional comparisons

The analytic initializer replaces the neural predictor with a low-rank formula for a grey-normalized temperature residual and $\log_{10} \kappa_R$. Column mass is then obtained by integrating $dm/d\tau = 1/\kappa_R$. The formula is fitted to the same atmosphere family but is evaluated only on the development sample. It provides a compact fitted comparison for profile accuracy and convergence. The hydrostatic grey-convective initializer provides the corresponding zero-fit physical comparison on the same development stars.

The analytic formula contains 2407 floating-point coefficients and fixed constants, fitted on 40911 atmospheres and checked on 10228 validation atmospheres. Its three temperature regimes use complete cubic functions of the stellar labels and five Chebyshev modes in depth. The formula is therefore much more compact than the neural model, but it remains a fitted description of the same atmosphere family.

The monotonic column-mass parameterization is also physically consequential. On held-out atmospheres it gives p_{95} errors of 3.74×10^{-3} in temperature and 1.52×10^{-2} dex in column mass, with 0 non-monotonic profiles. Directly predicting all depth points reduces these errors to 3.22×10^{-3} and 1.40×10^{-2} dex, but produces 3 inadmissible column-mass profiles. We therefore retain the monotonic construction.

Appendix F: Model configurations

Table F.1 summarizes the inputs, fitted quantities, and evaluation samples for each initialization method.

Appendix G: Reproducibility of reported quantities

All numerical values and table entries are generated from the saved numerical outputs by the same analysis used to construct the tables. A machine-readable record accompanying the manuscript links each reported quantity to its sample definition and numerical assumptions.

Table C.1. Convergence from the Paper I six-field initializer for different stellar-label distributions. The solver and initializer are identical in every row; only the sampling within the support of Sect. 4 differs. *Uniform box* draws each label uniformly over that support, which covers the box rather than the stellar locus and so includes combinations no real star occupies; *IID from corpus* resamples whole corpus rows with replacement, preserving the joint distribution the initializer was trained on; *label-box boundary* pins at least one label within 5% of a face of that support; *low gravity*, *metal poor* samples uniformly inside $\log g \in [0.7, 2.8]$, $[M/H] \in [-2.5, -0.5]$. Iteration counts include converged calculations only; q is the geometric-mean ratio of successive temperature residuals, with $q < 1$ indicating contraction.

Label sampling	N	Converged	Failed	Iterations	p_{90}	q	Non-monotonic
Uniform box	300	85.3%	14.7%	6.26	9	0.752	64.0%
IID from corpus	1000	99.2%	0.8%	5.86	8	0.644	56.6%
Label-box boundary	500	80.6%	19.4%	6.74	10	0.763	70.2%
Low gravity, metal poor	500	81.8%	18.2%	6.19	10	0.811	64.2%

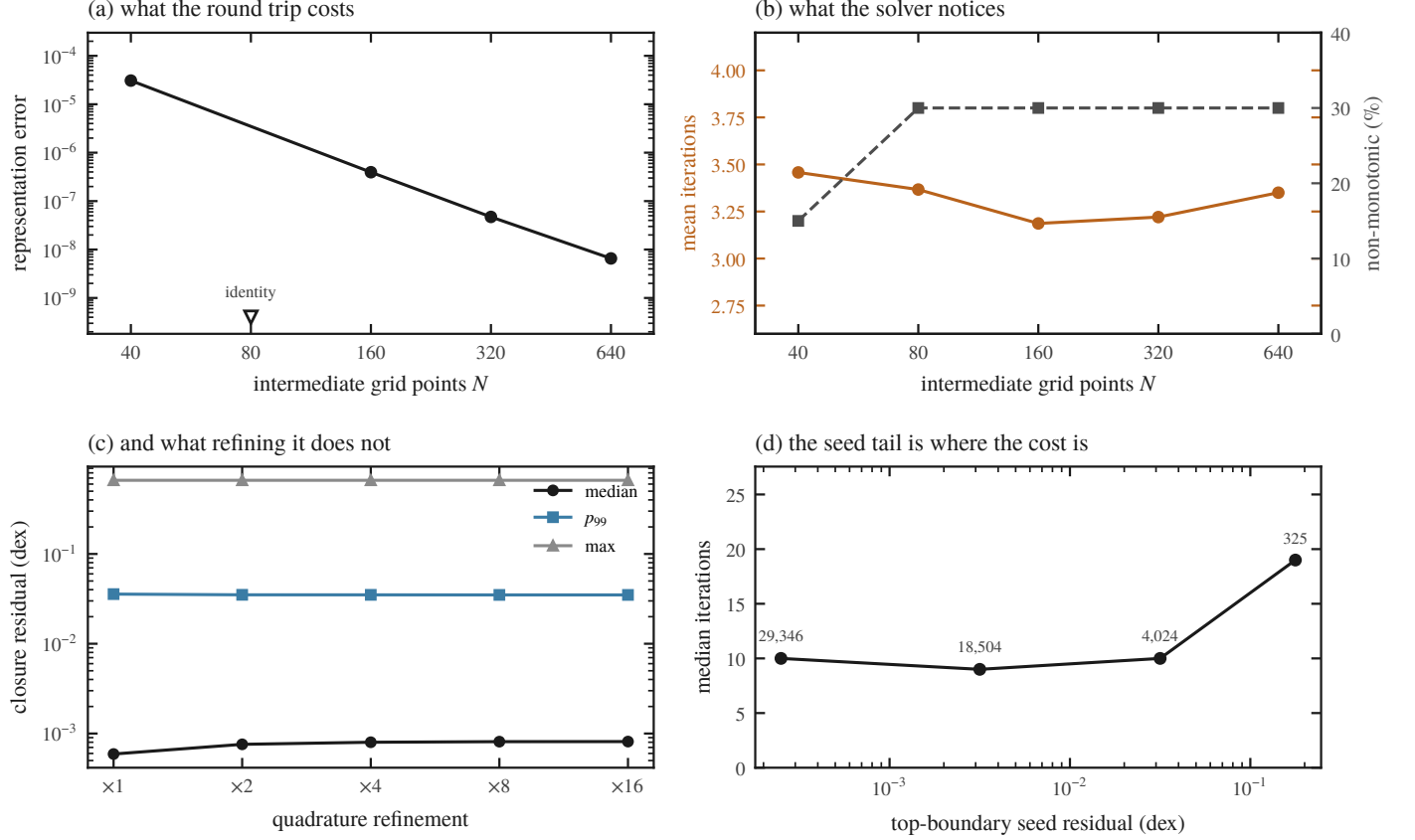


Fig. D.1. Dependence of reconstruction and convergence on numerical resolution. (a) Round-trip representation error as the intermediate grid is refined. (b) Mean solver iterations and non-monotonic fraction after physical reconstruction. (c) Median, p_{99} and maximum top-layer closure residual under quadrature refinement. (d) Solver iterations against the fixed surface-seed residual. Interior refinement changes the representation error but not the convergence behavior or the largest residuals. This control isolates representation and discretization error; it does not compare initialization methods.

Table E.1. Learned and analytic two-field initializers evaluated on the same 60 development stars. Profile errors are pooled over all 60×80 depth points. Iteration statistics include converged stars only; unsuccessful reconstruction and timeouts are counted as non-convergences. The parameter counts include independent floating-point coefficients and constants; integer basis indices are not counted. Solver calculations for the two methods were performed separately, and no end-to-end timing, spectral comparison, or independent-sample test is available for the analytic method.

Initializer	Fitted profiles	Float parameters	$T p_{95}$	$\log m p_{95}$	Converged	Mean iter.	Median
Learned two-field + physics	(m, T)	873 450	3.74×10^{-3}	1.52×10^{-2}	56/60	4.05	4.0
Fitted analytic formula	$(\delta_T, \log \kappa_R) \rightarrow m$	2 407	1.91×10^{-2}	8.76×10^{-2}	57/60	7.77	8.0

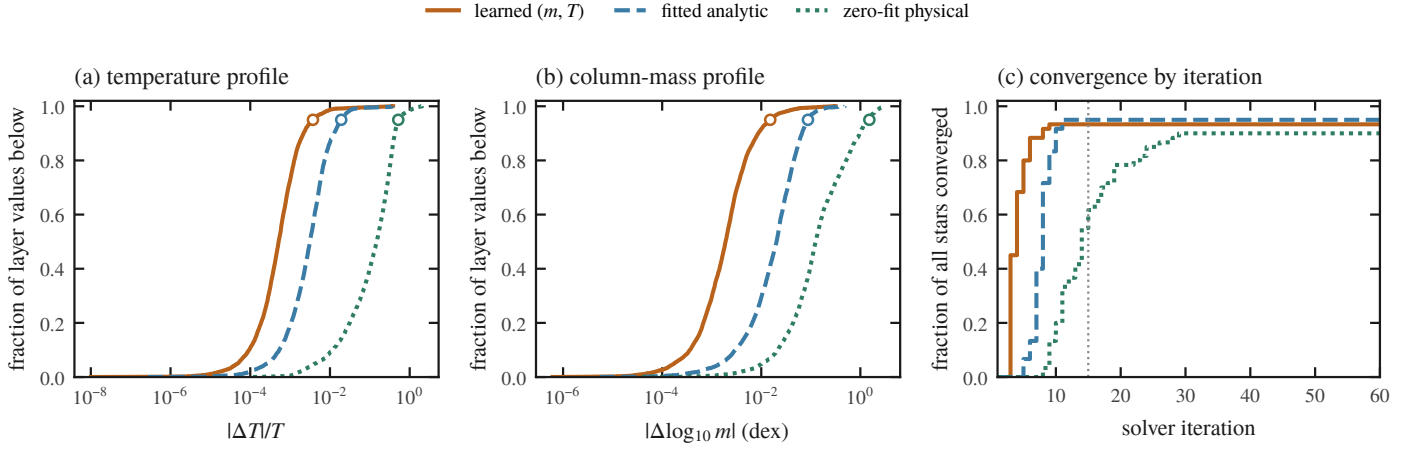


Fig. E.1. Same-star development comparison of the learned initializer, fitted analytic formula, and zero-fit hydrostatic grey–convective initializer. Panels (a) and (b) show cumulative layer-level temperature and column-mass seed errors, with open circles marking the p_{95} values. Panel (c) shows the fraction of all 60 stars converged by each iteration; stars that do not converge remain in the denominator. This separates the advantage of fitting an atmosphere family from the possibility of reaching the solver basin without fitted profiles.

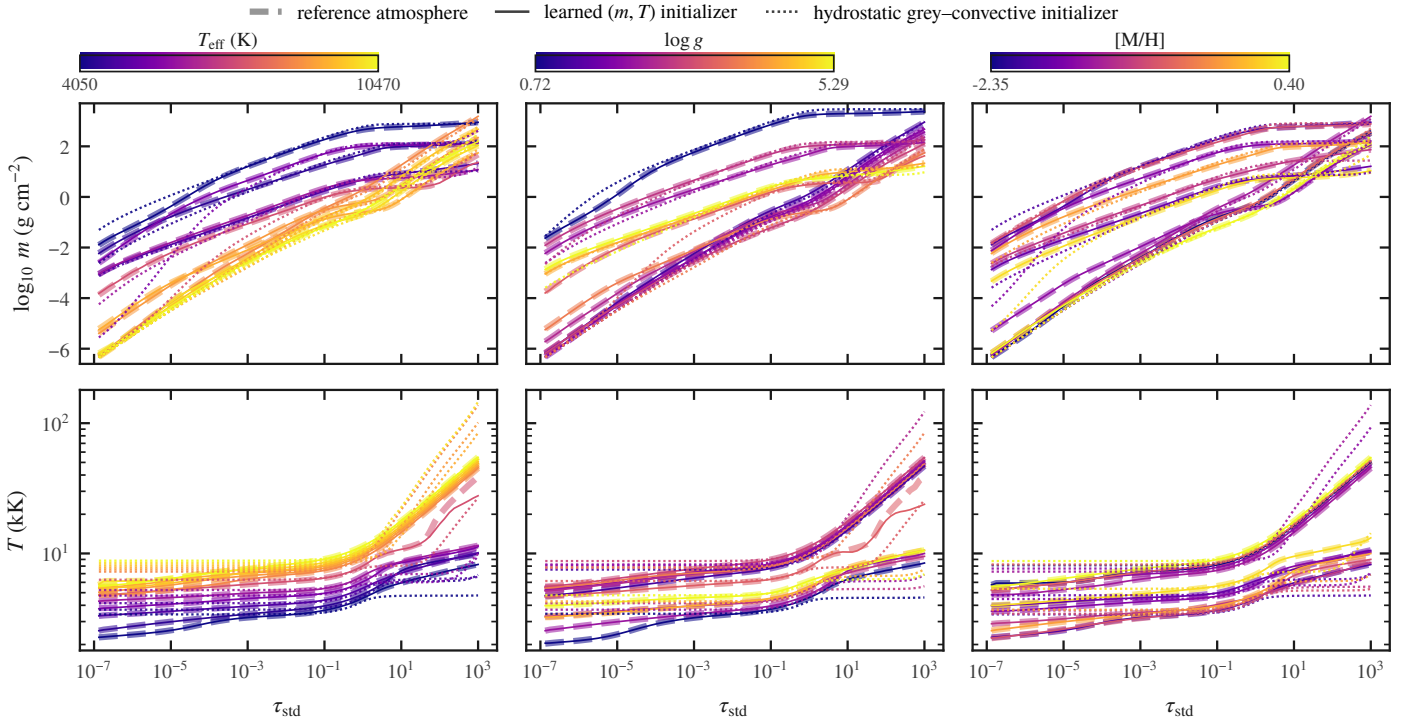


Fig. E.2. Two-field profiles on the development sample. Rows show column mass and temperature on the optical-depth grid; temperature is plotted logarithmically to retain the full physical-initializer range. Columns colour the same twelve representative stars by T_{eff} , $\log g$, and $[M/H]$. Long-dashed curves are the converged reference atmospheres, solid curves the learned seeds, and dotted curves the hydrostatic grey–convective seeds. The larger physical offsets are intentional approximations, not reconstruction residuals.

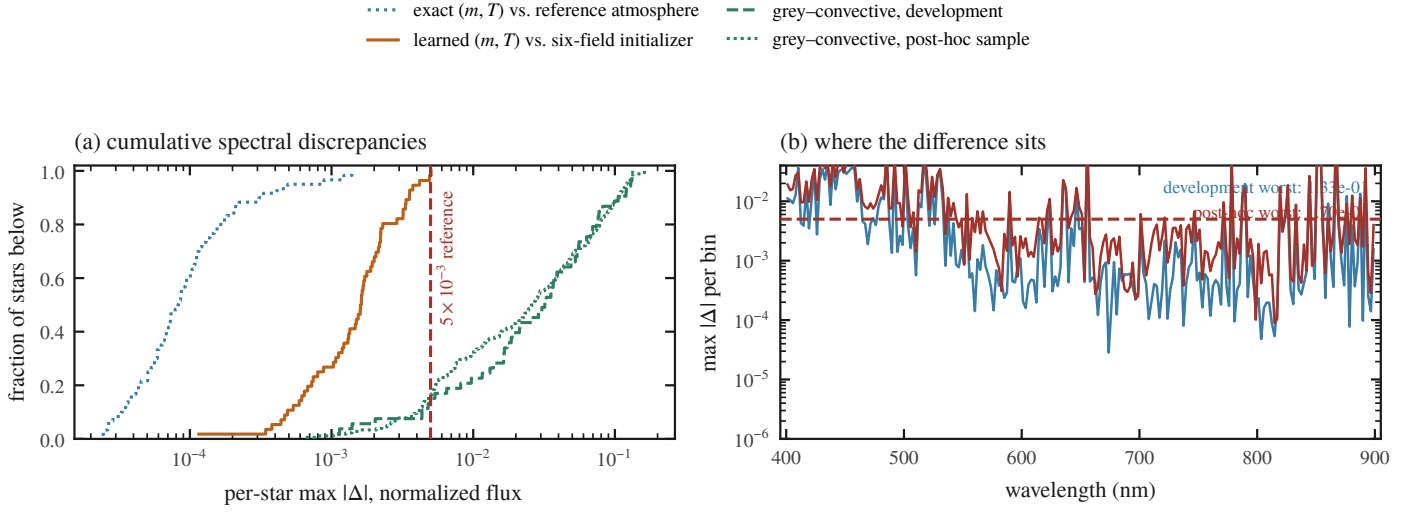


Fig. E.3. Spectral differences over 400–900 nm. (a) Cumulative per-star maximum normalized-flux differences for exact (m, T) against the reference six-field atmosphere, learned (m, T) against the six-field initializer, and the physical initializer on the development and 200-star samples. The vertical line marks the common 5.0×10^{-3} reference level. (b) Wavelength-binned maximum residuals for the worst paired physical-start spectrum in each sample. Curves include only stars for which both atmospheres converge; exclusion counts are given in Table 4.

Table F.1. Initialization methods and the evidence available for each. The independent-test model uses the radial-basis corrections defined in Sect. 5.3; the grey-convective construction contains no fitted atmospheric profiles.

Initialization	Input $\rightarrow$ state	Fitted information	Evidence
Six-field (Paper I)	5 labels $\rightarrow$ 6 fields, direct	Paper I; not retrained here	reference in Tables C.1, 3 and 6
Two-field emulator (this work)	5 labels $\rightarrow$ ($\log_{10} m, \log_{10} T$), monotone increments of Eq. (10)	4×512 SiLU network; 41 867 training and 10 272 validation atmospheres; development stars excluded	development comparisons in Tables 2, 3
Fitted analytic formula (this work)	5 labels $\rightarrow$ (T, κ_R); m from $dm/d\tau = 1/\kappa_R$	40 911/10 228 train/validation rows; three T_{eff} regimes, cubic labels and five Chebyshev depth modes; 2 407 floating-point coefficients and fixed constants	profile and development-sample solver comparison in Table E.1; no spectral or independent test
Grey-convective physical model	5 labels $\rightarrow m_{\text{grey}}, T_{\text{conv}}, \kappa_R$; no post-convection mass reintegration	no fitted profiles; approximate opacity, hydrostatic $P = g m_{\text{grey}}$ and Saha-aware convective temperature	development-sample comparison in Fig. 6; convergence in Table 4; post-hoc 200-star comparison and paired spectra
Independent-test two-field model	two-field base + two label-local radial-basis corrections	base retrained without the development or independent-test rows; a three-centre $\sigma = 0.30$ one-step correction and a 20-centre $\sigma = 0.08$ profile correction, all fixed before testing	independent test in Table 6 and Fig. 8